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Acoustic state detection of microbubble emission boiling using a deep neural network based on cepstrum analysis

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ABSTRACT

Microbubble emission boiling (MEB) is a cooling technology in which the heat flux can potentially exceed the critical heat flux (CHF). Reliable predictions of the occurrence of MEB are necessary to achieve stable MEB and to induce it under actual environment conditions. In this study, we developed a method based on deep learning with boiling sound to predict the boiling state of the interval before the low-heat-flux level reaches MEB. The boiling sound was acquired by a hydrophone, and the sound was adopted to machine learning algorithms, which were subsequently applied to classification and regression models. The feature extraction algorithms for the boiling sounds were spectrum or cepstrum methods. Both methods were comparatively investigated in terms of the machine learning accuracy. As a result, in the case of the cepstrum method as the feature extraction, the accuracy was improved. In particular, we found that the regression model demonstrated substantially better accuracy than the classification model. In addition, accurate predictions were possible even when the degree of subcooling was changed.

1. Introduction

Technological advances and further downsizing of electronic equipment has led to an increase in heat generation density (Murshed and De Castro, 2017). Boiling cooling technology with high cooling performance is attracting attention as a method to prevent problems related to electronic equipment that generates substantial heat (Fan and Duan, 2020). One of the associated technologies is microbubble emission boiling (MEB) (Inada et al., 1981), which is capable of high-heat-flux removal greater than 1000 W/cm^2 (Suzuki et al., 2005). Since onset of MEB is observed at subcooled pool boiling, the condensation of vapor bubbles is widely studied (Tang et al., 2022). At the time of MEB inception, the wall superheating degree might exceed the critical heat flux (CHF) (Unno et al., 2022). If such an MEB state can be precisely predicted, then MEB can be applied in actual cooling equipment.

With the rapid development of both computer performance and machine learning (ML) algorithms in recent years, deep learning is being applied in various research fields. Examples of applications of ML include image recognition (Krizhevsky et al., 2017), super-resolution image technology (Chen et al., 2017), and image noise reduction (Lehtinen et al., 2018). ML is currently being used in thermal engineering, including applications such heat transfer material informatics (Ju et al., 2017), thermofluid analysis (Fukami et al., 2019; 2020), and combustion flow (Kobayashi et al., 2019). It is also being applied to multiphase flows. The literature includes investigations of flow-pattern discrimination of gas–liquid two-phase flow by unsupervised learning (Julia et al., 2011), two-phase flow of flow pattern identification (Poletaev et al., 2020; Nie et al., 2022), sound recognition of gas–liquid two-phase flow based on ML (Zhang et al., 2021), acoustic sensing of bubble-jet flow and identification of gas leak diameters (Mikami et al., 2023), and the application of ML to boiling data of infrared thermal images (Ravichandran et al., 2021). In addition, several studies of ML using boiling sound (Barathula and Srinivasan, 2022) have been reported (Sinha et al., 2021; Ueki and Ara, 2021; Barathula et al., 2023; Dunlap et al., 2023). It is because the acoustic sensing has several advantages of fast-response, in-site measurement in high spatial-temporal resolutions. The most powerful advantage is that the acoustic sensing works even in optically invisible environment. Acoustics of the MEB had been experimentally investigated so far (Kubo et al., 1993; Suzuki and Inagaki, 2005; Tang et al., 2018; Kawakami et al., 2021; Kobayashi et al., 2022). Yet, relation between the acoustics and heat transfer

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Nomenclature

k	thermal conductivity (W/mK)
q	heat flux (W/m^2)
s	distance (m)
T	temperature (°C)
T_{sat}	saturation temperature
T_w	wall temperature
ΔT_{sat}	wall superheat (K), T_w - T_{sat}
ΔT_{sub}	liquid subcooling (K)
U	uncertainty

Subscripts

w	wall
sat	saturation
sub	subcooling

characteristic had not been fully elucidated.

Therefore, in the present study, we investigated the prediction precision of heat flux and wall superheat in the boiling state including an onset of MEB in detail using deep learning of the boiling sound to predict the subcooled boiling state that includes the MEB. In addition, we used cepstrum analysis (Bogert et al., 1963), which is frequently utilized in speech recognition and machine diagnostics (Randall, 2017). The cepstrum analysis is an effective signal processing algorithm. However, it has not ever been applied to the boiling acoustics. In the present study, discrete cepstrum signals of the boiling sound were also used in the ML. Both a regression model and classification model were applied, and the effectiveness as the feature extraction was comparatively investigated.

2. Experimental apparatus and methodology

2.1. Experimental apparatus for subcooled boiling

Fig. 1 shows a schematic of the experiment system for subcooled pool boiling at atmospheric pressure. This system is similar to that reported elsewhere (Ono et al., 2021). The side walls of the vessel were made of glass plates. The vessel was 150 mm wide, 150 mm long, and 200 mm high. The top plate and a bottom plate were made of stainless steel. In the present study, the working fluid for the boiling test was distilled water, which was previously boiled to remove dissolved gases. The water level was 150 mm from the heat transfer surface. The degree of liquid subcooling was controlled at 40, 50, or 60 K using a heat exchanger and auxiliary heaters inside the vessel. A chiller was used to control the temperature of coolant circulating through the heat exchanger. The temperature of the coolant was 15°C. A stirrer weakly agitated the water to make the liquid subcooling distribution as uniform as possible. A ⌀9 mm oxygen-free copper surface was used for the heat transfer surface. Four cartridge heaters were inserted at the bottom of the copper heating block. In the present study, the input power of the heaters was varied by changing the applied voltage using a voltage converter. The input power was increased in 5 V steps, and the experiment was completed when MEB occurred after a steady state had been established. The heating block was covered with a thermal insulator to prevent heat leaks. The heating block included a cylindrical section, and three thermocouples (K-type, ⌀0.5 mm, Class 1) were inserted with even, 3 mm spacing from the top of the copper block. The tip of the thermocouples was located at the center axis of the cylindrical section. The temperature of the water inside the vessel was also measured with another thermocouple. The temperatures measured by the four thermocouples at a steady state via a data logger were averaged for 1 min. Using the obtained temperature data, the heat flux and the surface temperature were estimated using Fourier's law because the temperature gradient in the cylindrical part was reasonably assumed to be one-dimensional heat conduction. Actually, the coefficient of determination (R^2) was greater than 0.995 in all the data, as shown in Fig. 2. Therefore, we assume the heat loss was ignorable in this study. The slope and intercept of the temperature distribution were determined by the least-squares method. A hydrophone immersed in the water was used to measure boiling sounds. The sampling rate was 48 kHz. In addition, a high-speed camera and an LED light were used to observe the behavior of boiling bubbles. The camera and light were synchronized with each other at 10 kHz through a signal generator. The capture size of the camera image was 112 × 112 pixels.

The uncertainties U of the heat flux q and the wall superheat ΔT_{sat} were estimated as follows:

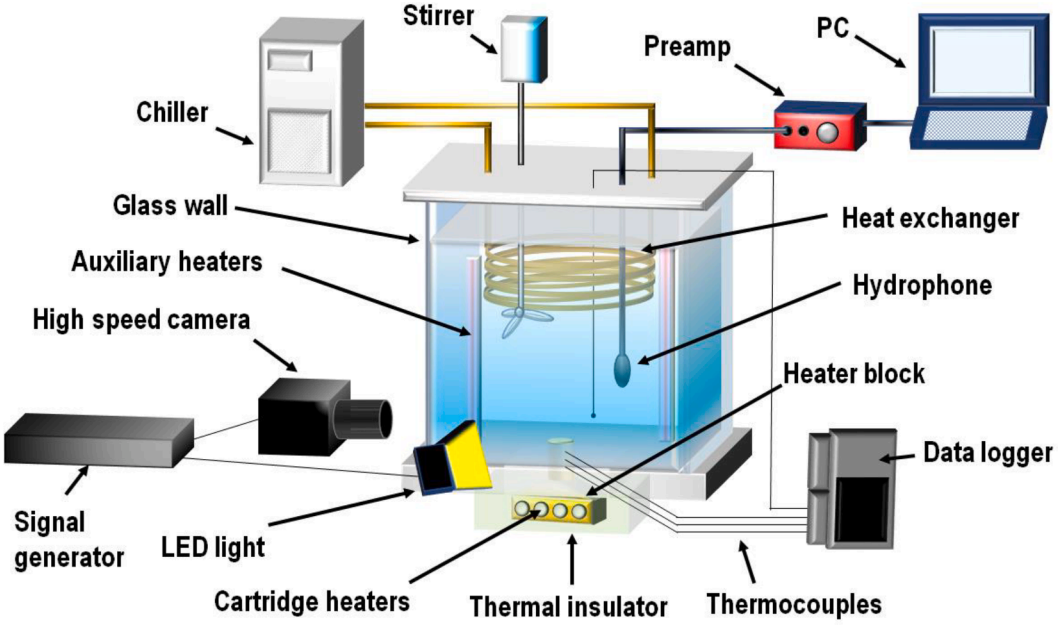


Fig. 1. Experimental setup for subcooled pool boiling experiments under atmospheric conditions using a copper heating block.

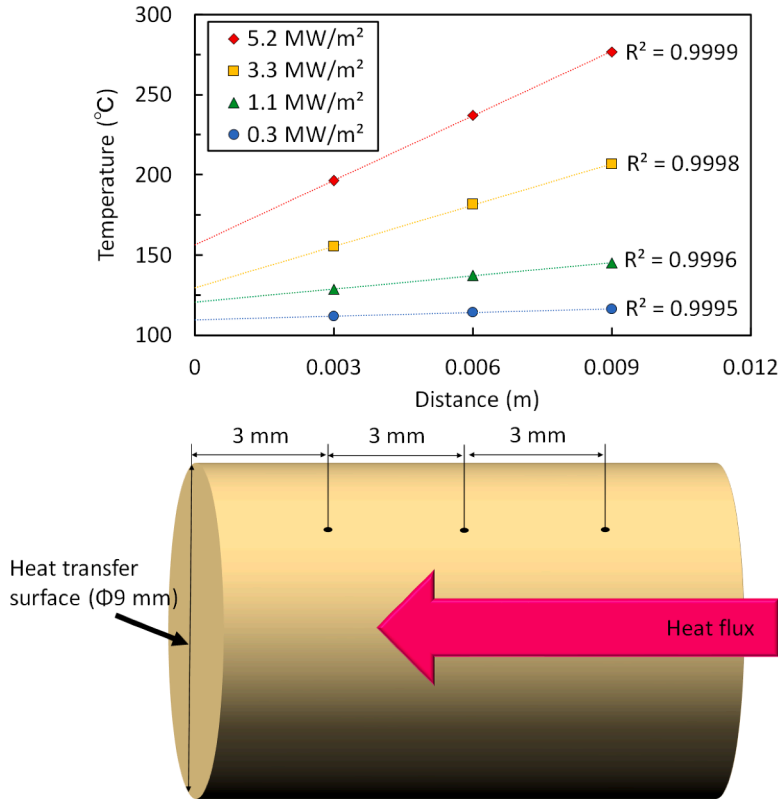


Fig. 2. Details of the cylindrical part of the copper heating block containing three thermocouples.

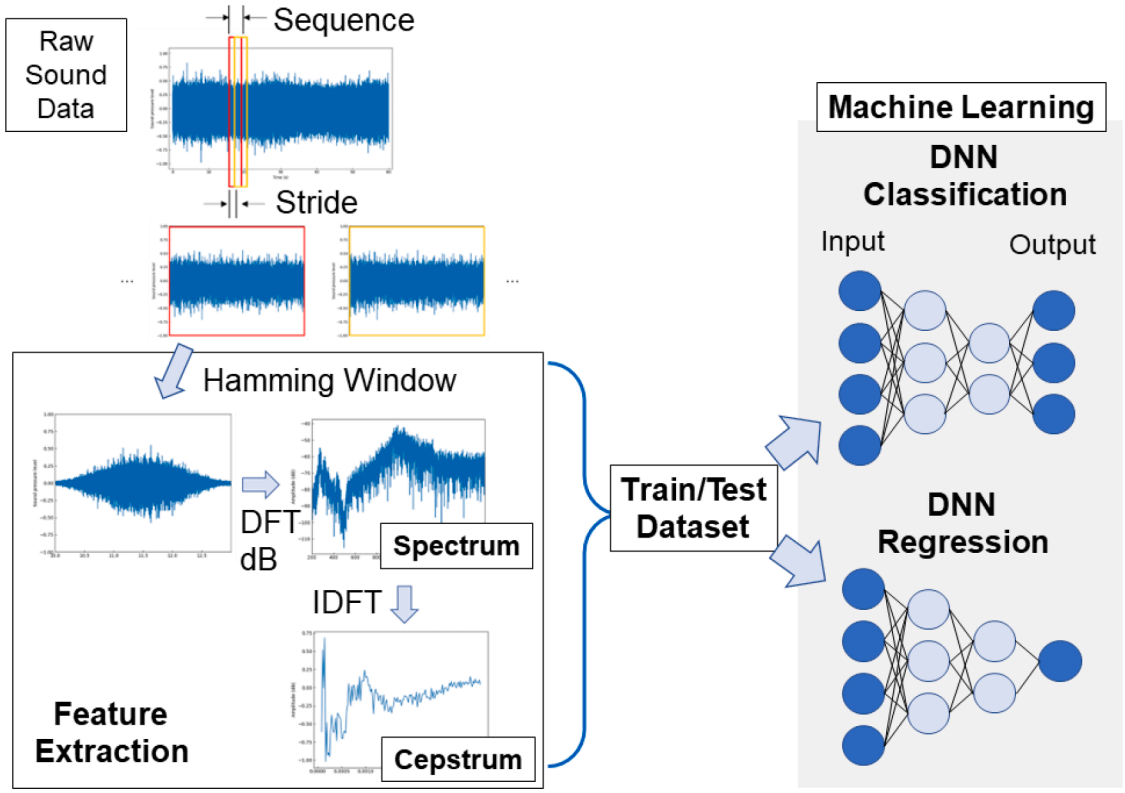


Fig. 3. Illustration of the machine learning model framework for acoustic-based quantification of boiling heat flux and wall superheat.

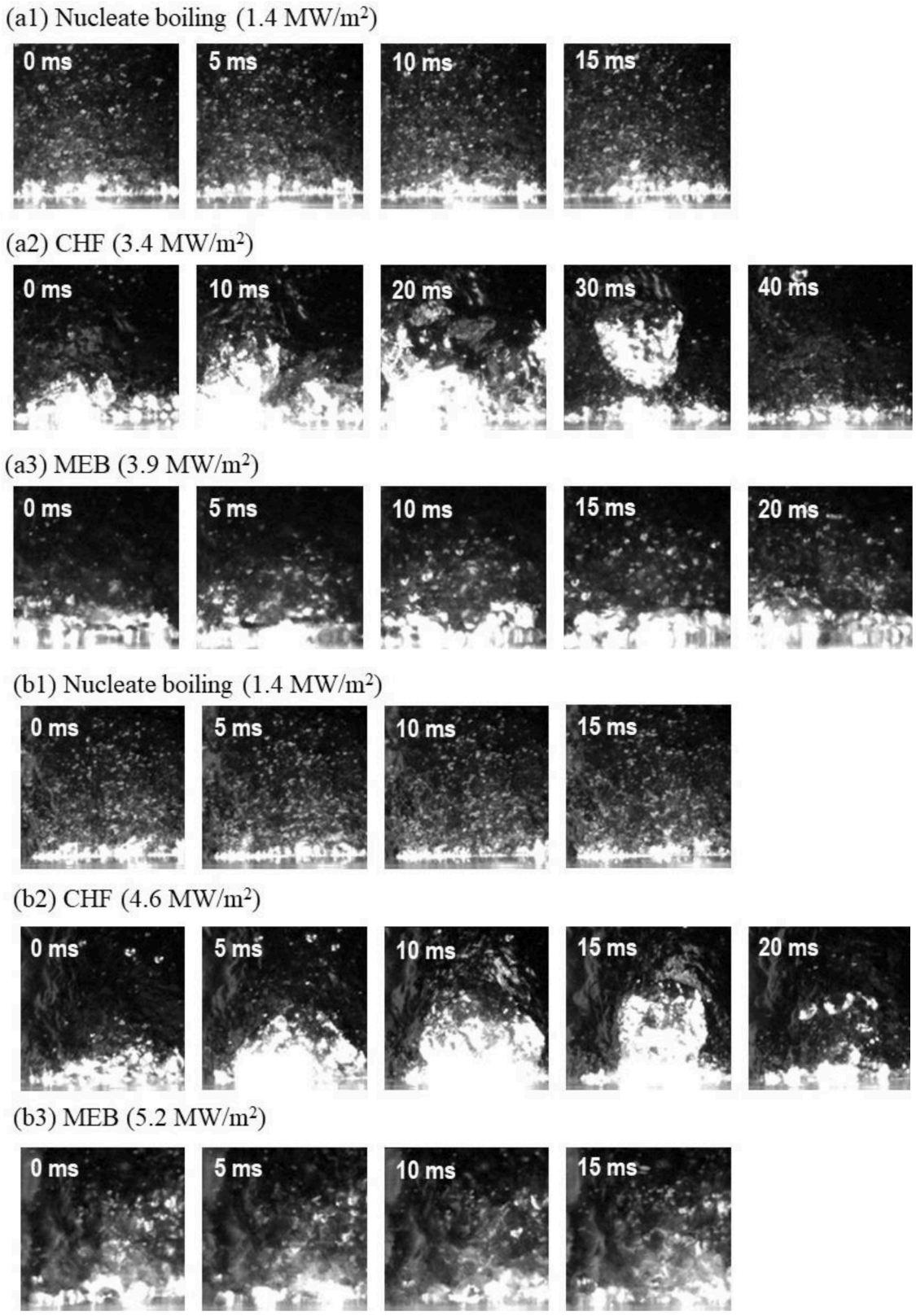


Fig. 4. Typical boiling bubbles at various degrees of liquid subcooling ΔT_{sub}: (a) 40 K, (b) 50 K, and (c) 60 K.

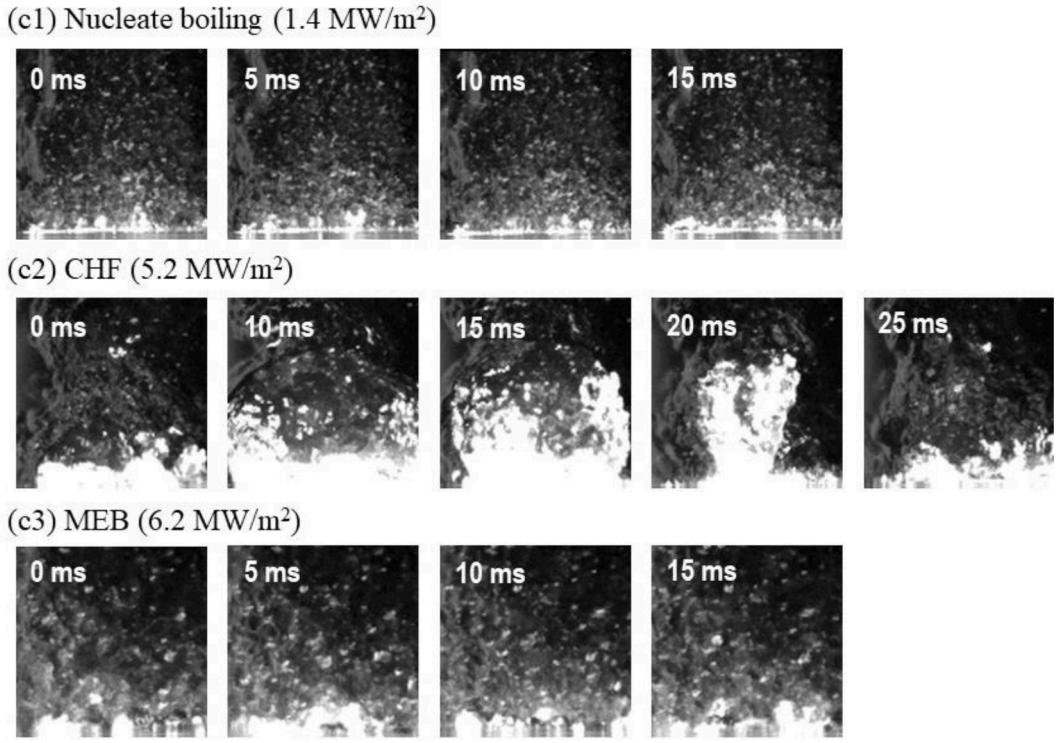


Fig. 4. (*continued*).

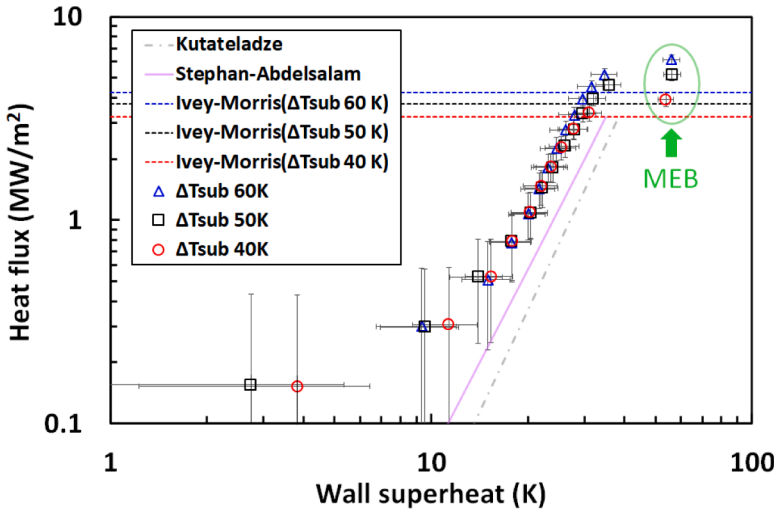


Fig. 5. The obtained boiling curves as a function of the liquid subcooling ΔT_{sub}.

$$q = k\frac{T_2 - T_1}{s}, \tag{1}$$

$$\Delta T_{sat} = T_w - T_{sat} = T_1 - \frac{qs}{k} - T_{sat}, \tag{2}$$

$$U(q) = \sqrt{\left(\frac{\partial q}{\partial T_1}U(T_1)\right)^2 + \left(\frac{\partial q}{\partial T_2}U(T_2)\right)^2 + \left(\frac{\partial q}{\partial s}U(s)\right)^2}, \tag{3}$$

$$U(\Delta T_{sat}) = \sqrt{\left(\frac{\partial \Delta T_{sat}}{\partial T_1}U(T_1)\right)^2 + \left(\frac{\partial \Delta T_{sat}}{\partial q}U(q)\right)^2 + \left(\frac{\partial \Delta T_{sat}}{\partial s}U(s)\right)^2}. \tag{4}$$

Here, because the vessel was opened to the air, the saturation temperature of water T_{sat} was considered to be 100°C. T_1 and T_2 are the temperatures at the positions of 3 and 6 mm, respectively, and s is the distance between the thermocouples in the cylindrical section (0.003 m). The fitting curve for copper thermal conductivity k was obtained from previously reported data (White and Minges, 1997). At the maximum heat flux of 6.2 MW/m^2 in the present study, the estimated uncertainties of heat flux $U(q)$ and wall superheat $U(\Delta T_{sat})$ were 0.5 MW/m^2 and 4 K, respectively. The uncertainty values are depicted as error bars in figures containing experimental results.

2.2. Model architecture for acoustic sequence regression

2.2.1. Data sampling

The sampling rate was 48 kHz, the bit depth was F32 bit, and the boiling sound was recorded for each point of the boiling curve for 60 s as

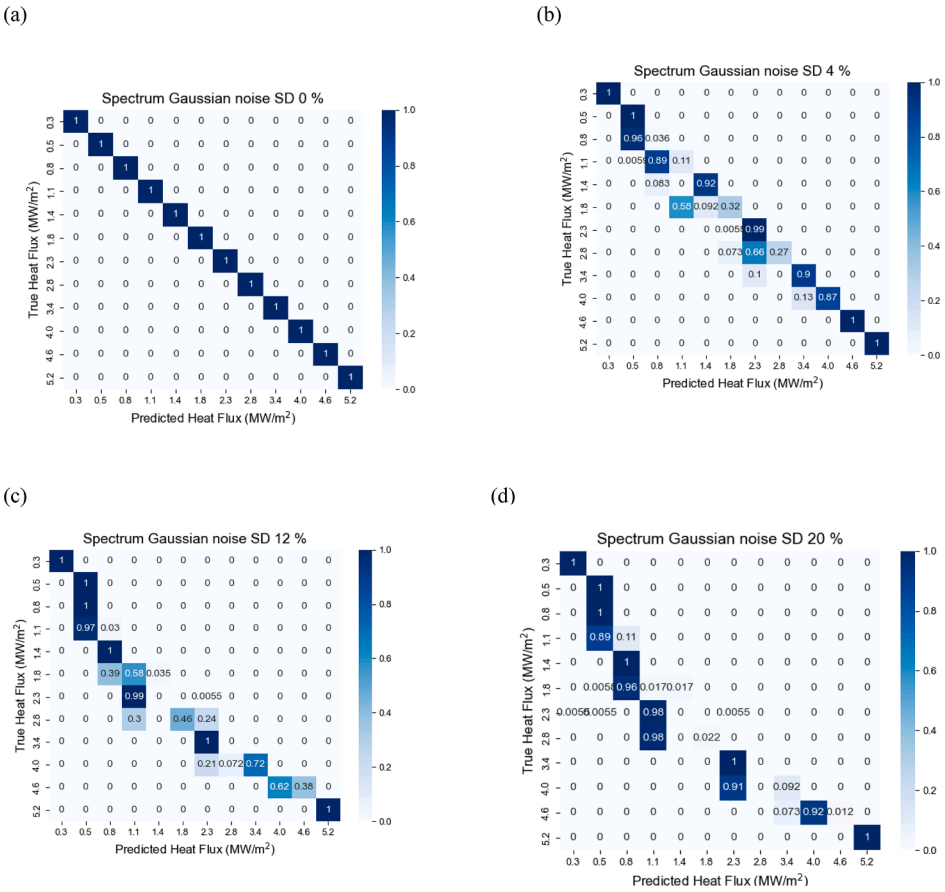


Fig. 6. Confusion matrix of DNN classification for the spectrum dataset with various SDs at $\Delta T_{sub} = 50$ K: (a) 0%, (b) 4%, (c) 12%, and (d) 20%.

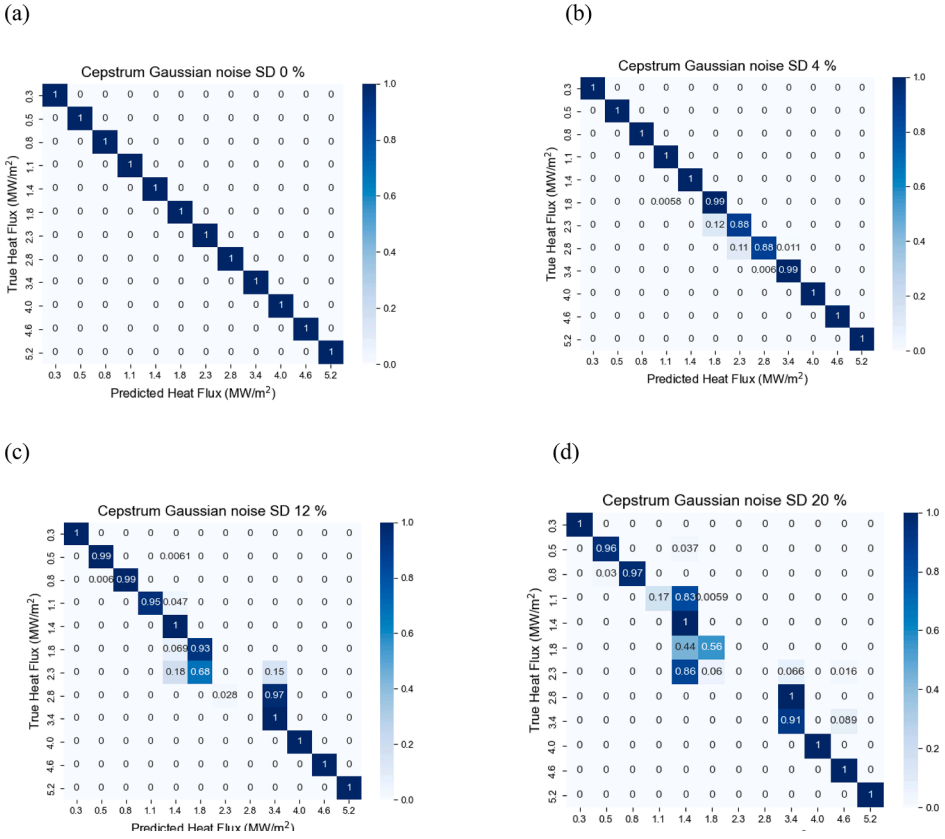
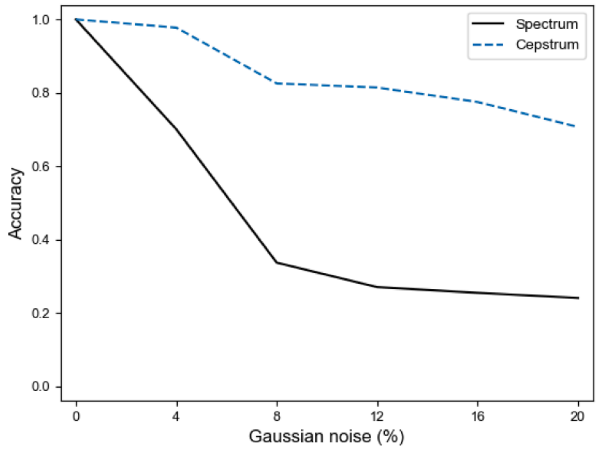


Fig. 7. Confusion matrix of DNN classification for the cepstrum dataset with various SDs at $\Delta T_{sub} = 50$ K: (a) 0%, (b) 4%, (c) 12%, and (d) 20%.

(a)



(b)

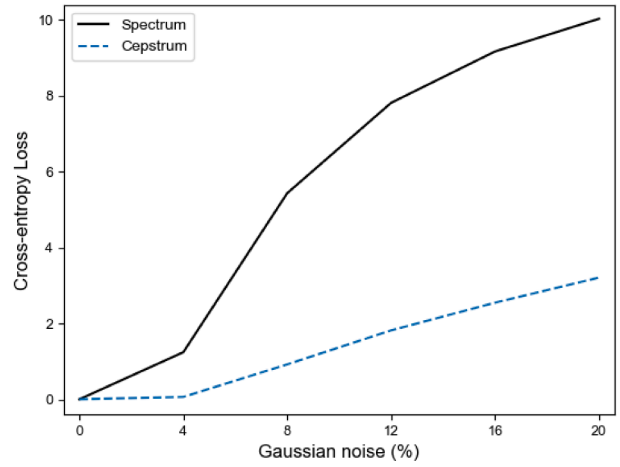


Fig. 8. DNN classification model with Gaussian noise at $\Delta T_{sub} = 50$ K: (a) plots of accuracy and (b) cross-entropy loss.

a WAV file. In this study, the sound was acquired at above 0.3 MW/m^2 because the boiling state was developed over the heat transfer surface. The sound data were divided by the stride in sequence length as 0.1 s for 3 s. The amplifier (Focusrite Clarett+ 2 Pre USB) had a maximum sampling rate of 192 kHz.

2.2.2. Feature extraction

The data processing for the measured sound source is shown in Fig. 3. A Hamming window was applied to the extracted data. A fast Fourier transform (FFT) algorithm was then applied to the data, and a spectrum was obtained. A common logarithm was applied to the obtained spectrum 20 times, and the spectrum was converted to decibels (dB). The dB data in the spectrum model were cut and brought within the 200–1800 Hz band for ML. We applied an inverse fast Fourier transform (IFFT) algorithm to the dB data to obtain the cepstrum. We abandoned four points at the beginning of the cepstrum model and carried out ML using 160 points beginning from the fifth point. The libraries used for the FFT and IFFT were scipy.fft.fft and scipy.fft.ifft, respectively; the version was scipy 1.7.1.

2.2.3. Classification and regression models

The Tensor Flow and Keras ML libraries were used, along with the Anaconda3 2021.11 and Python 3.9.7 program development environments. We employed a sequential model with supervised learning. The classification model was constructed using heat flux as a label to detect profiles on the boiling curves. The label numbers of the subcooling degrees of 40, 50, and 60 K were 10, 12, and 13, respectively.

In the network architecture of the spectrum classification model, the number of inputs was 4800 and the number of outputs of layers 1–4 resulted in five output levels, matching the number of the labels with 2400, 1200, 600, and 300 outputs in all five levels of the combination neural networks. The five levels used the softmax function as an activation function for layers 1–4 via a rectified linear unit (ReLU) function. In the network of the cepstrum classification model, the number of inputs was 160 and the number of outputs of layers 1–4 resulted in five output levels, matching the number of the labels with 80, 40, 20, and 10 in all five levels of each combination neural network. The five levels used the softmax function as an activation function for layers 1–4 layer via an ReLU function. The learning rate was set at 0.001. The optimization algorithm was cross entropy. The Adam algorithm was used to optimize the loss function in the classification models.

We constructed regression models for heat flux and wall heat. In the network of the spectrum regression model, the number of inputs was assumed to be 4800, the number of outputs of layers 1–5 was 2400, 1200, 600, 300, and 1 in all five levels of each combination neural network. An ReLU function was used as an activation function for layers 1–4. In the network of the cepstrum regression model, the number of inputs was assumed to be 160 and the number of outputs of layers 1–5 was 80, 40, 20, 10, and 1 in all five levels of each combination neural network. An ReLU function was used as an activation function for layers 1–4. The learning rate was set at 0.001 in mean squared error (MSE) levels, and the Adam algorithm was used to optimize the loss function in these regression models.

To inspect the performance of the learned model, we prepared sound data with Gaussian noise added. With respect to the standard deviation (SD) of the Gaussian noise, the ratio for the SD of the sound-pressure level in the original sound data was 4%, 8%, 12%, 16%, and 20%. The spectrum and cepstrum were obtained for the data to which this noise was added equally; the spectrum and cepstrum were then input into the model that had been learned. We used Microsoft Windows 10 Enterprise 64-bit OS, an NVIDIA GeForce RTX 3060 12GB GPU, and an Intel Core i5-10400 2.9GHz CPU to process the data.

3. Results and discussion

Fig. 4 shows boiling bubbles in the nucleate boiling regime, the CHF, and the MEB at various degrees of liquid subcooling. Fig. 5 depicts the obtained boiling curves, along with the predictions of boiling heat transfer in the nucleate boiling regime (Kutateladze, 1952; Stephan and Abdelsalam, 1980). Despite the different subcooling, no significant difference was observed in the boiling behavior in the nucleate boiling regime. Therefore, the boiling curves in Fig. 5 correspond well to each other in this regime. However, coalescing bubbles were generated on the heat-transfer surface at the CHF. The Ivey–Morris correlation is used to

predict the CHF in subcooled pool boiling (Ivey and Morris, 1966), and a higher CHF is known to be observed with greater liquid subcooling. In the present study, the obtained CHFs were 3.4, 4.6, and 5.2 MW/m^2 at liquid subcooling degrees of 40, 50, and 60 K, respectively. After the transition to MEB, large coalescing bubbles were not generated and many microbubbles were emitted from the heat transfer surface.

ML was used for three cases of calculations of ΔT_{sub}; however, the results for $\Delta T_{sub} = 40$ and 60 K are not shown. Only the results for $\Delta T_{sub} = 50$ K are discussed here because the tendency of the results does not change. Fig. 6 shows confusion matrices of the deep neural network (DNN) classification for the spectrum dataset with various SDs. The classification performance worsens from the low-heat-flux side because of increasing noise. Confusion matrices of the DNN classification for the cepstrum dataset with various SDs are shown in Fig. 7. The classification performance of the cepstrum model decreases with increasing noise near the heat-flux bunch of the medium domain. In the case of 20% noise, the learning rate is maintained in the high-heat-flux regime.

The accuracy and cross-entropy loss distributions for the DNN classification model with Gaussian noise are shown in Fig. 8. In the accuracy and cross-entropy loss distributions, the cepstrum model shows greater noise tolerance than the spectrum model. Fig. 9 shows the heat flux predicted by the regression DNN model for the spectrum dataset with various SDs. In the absence of noise, the predictions agree well with the experimental values. However, the precision deteriorates in the intermediate-heat-flux region as noise increases. The heat flux predicted by the regression DNN model for the cepstrum dataset with various SDs is shown in Fig. 10. The predictions show good agreement with the experimental values in the absence of noise. As noise increases, the predicted values tend to be too small in the intermediate-heat-flux region but well match the experimental values in the other heat-flux regions. Fig. 11 shows the wall superheat predicted by the regression DNN model for the spectrum dataset with various SDs. With increasing noise, the predicted wall superheat values well match the experimental results. Fig. 12 shows the wall superheat predicted by the regression DNN model for the cepstrum dataset with various SDs. In this case, the predictive precision in certain ranges is well maintained even when the noise is increased.

These results indicate that predictions based on the cestrum dataset show better predictive precision for both the heat flux and the wall superheat compared with the predictions based on the spectrum dataset. To quantify this observation, we calculated another quantity. Fig. 13 shows the coefficient of determination (R^2) for the heat-flux distribution and the wall-superheat distribution for the DNN regression model with Gaussian noise added. Even in the case of 20% noise, the R^2 of the predictions of the cepstrum model does not deteriorate; by contrast, the R^2 of the predictions of the spectrum model deteriorate sharply beginning at 4% noise. Fig. 14 shows the mean absolute error (MAE) for the DNN regression models with Gaussian noise. Even when the noise increases, the model based on the cepstrum dataset can suppress the error.

4. Conclusions

Three cases of subcooling degrees were investigated, and the boiling sound in the high-heat-flux area before MEB occurred was acquired and used for ML. The predictions showed high precision when no noise was added, irrespective of whether the classification was based on the regression, spectrum, or cepstrum model. Deterioration of the predictive precision was remarkable in the model based on spectrum data with noise. By contrast, the model based on cepstrum data maintain constant precision. Cepstrum data do not include information related to small

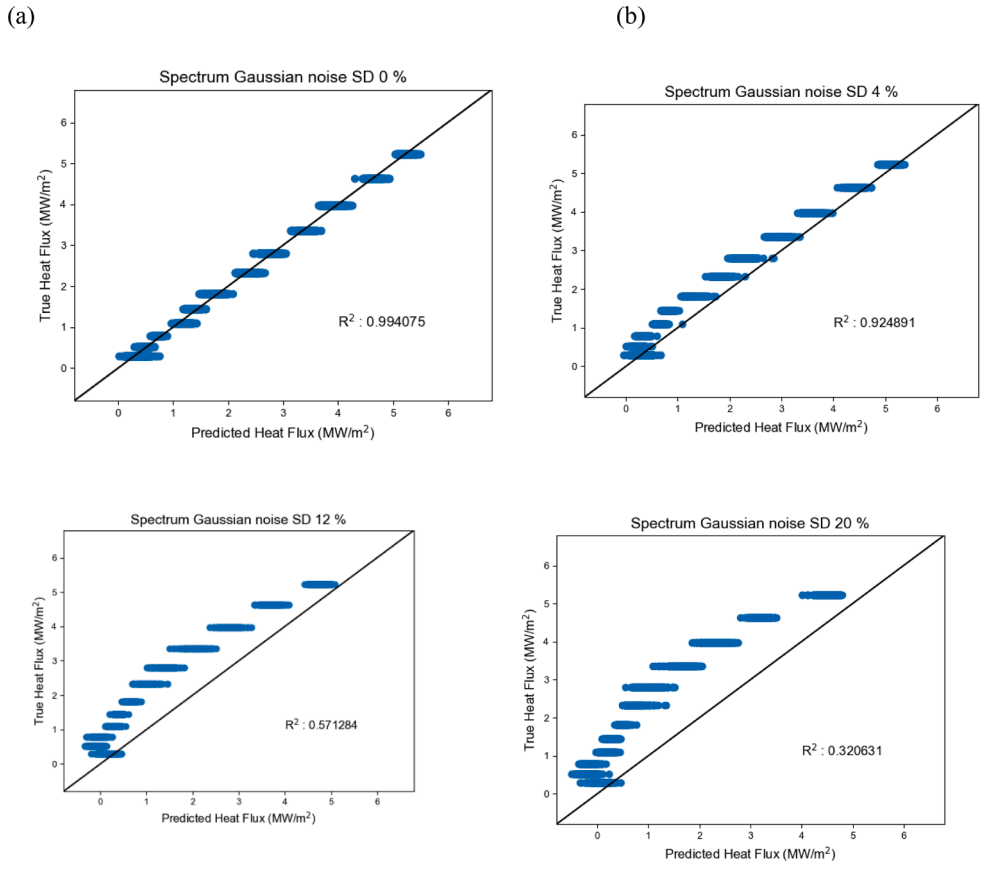


Fig. 9. Heat flux predicted by the regression DNN model for the spectrum dataset with various SDs at $\Delta T_{sub} = 50$ K: (a) 0%, (b) 4%, (c) 12%, and (d) 20%.

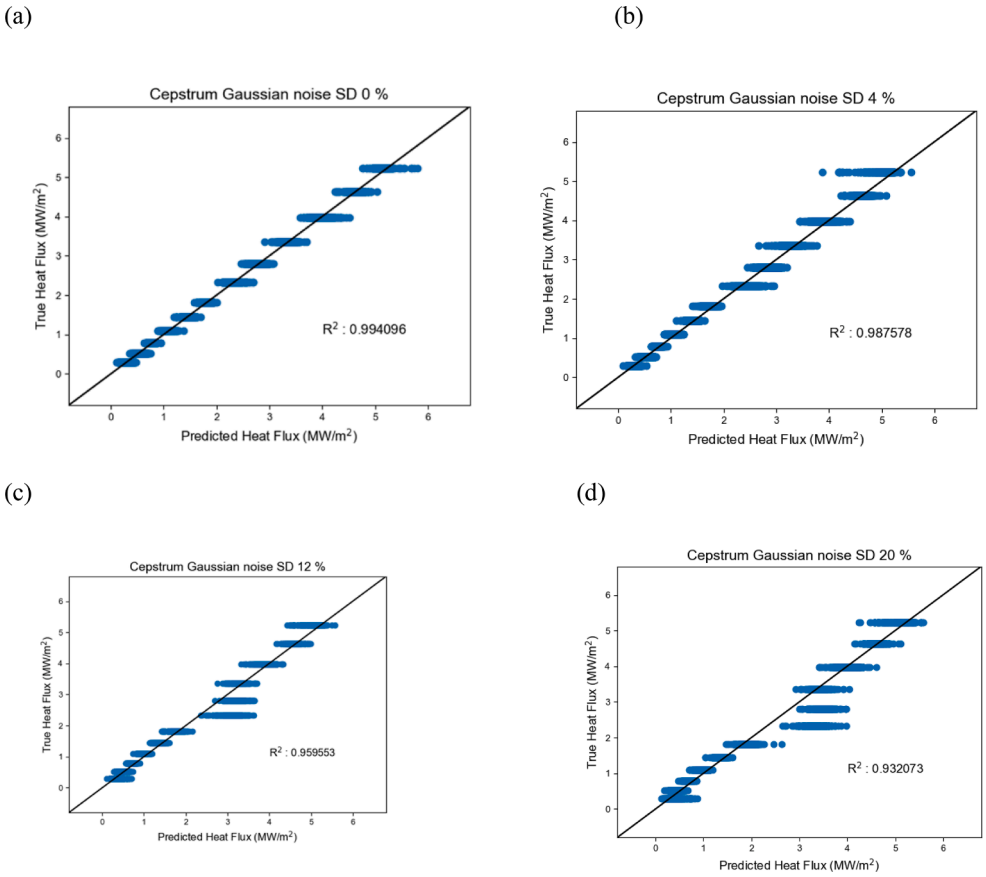


Fig. 10. Heat flux predicted by the regression DNN model for the cepstrum dataset with various SDs at $\Delta T_{\mathrm{sub}} = 50$ K: (a) 0%, (b) 4%, (c) 12%, (d) 20%.

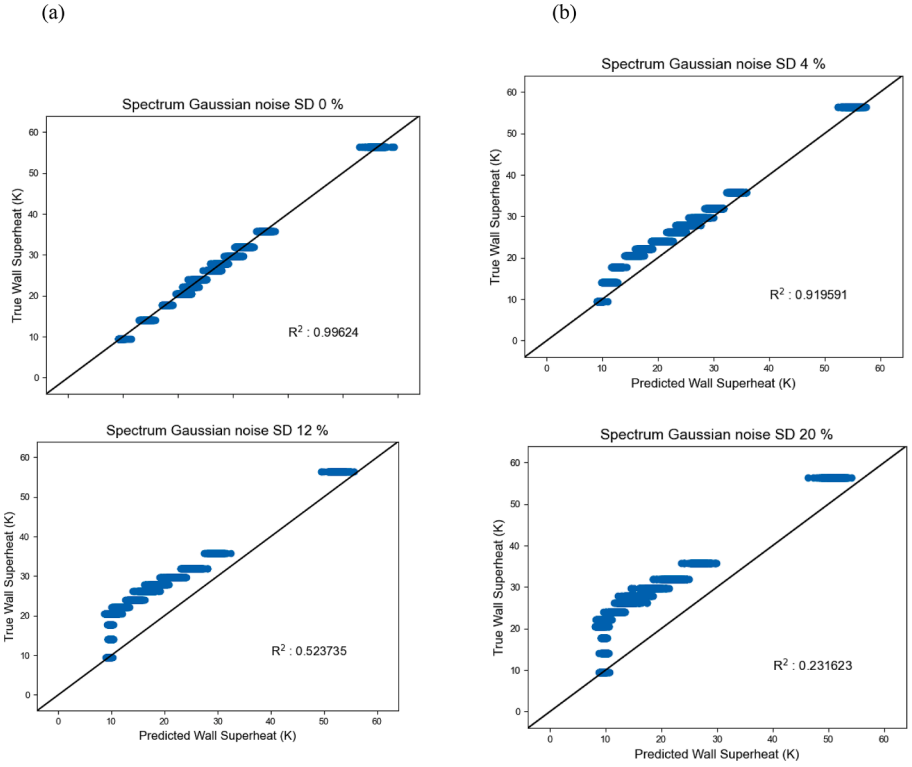


Fig. 11. Wall superheat predicted by the regression DNN model for the spectrum dataset with various SDs at $\Delta T_{\mathrm{sub}} = 50$ K: (a) 0%, (b) 4%, (c) 12%, and (d) 20%.

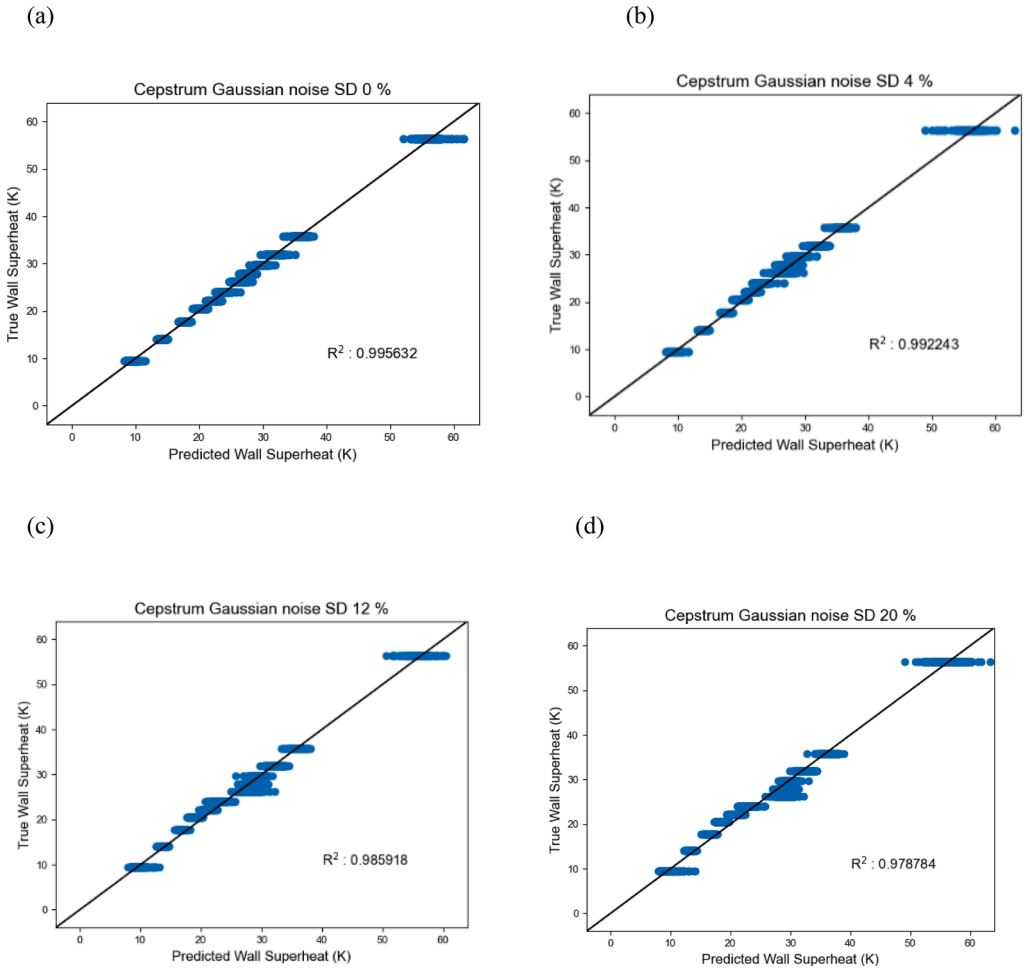


Fig. 12. Wall superheat predicted by the regression DNN model for the cepstrum dataset with various SD at $\Delta T_{sub} = 50$ K: (a) 0%, (b) 4%, (c) 12%, and (d) 20%.

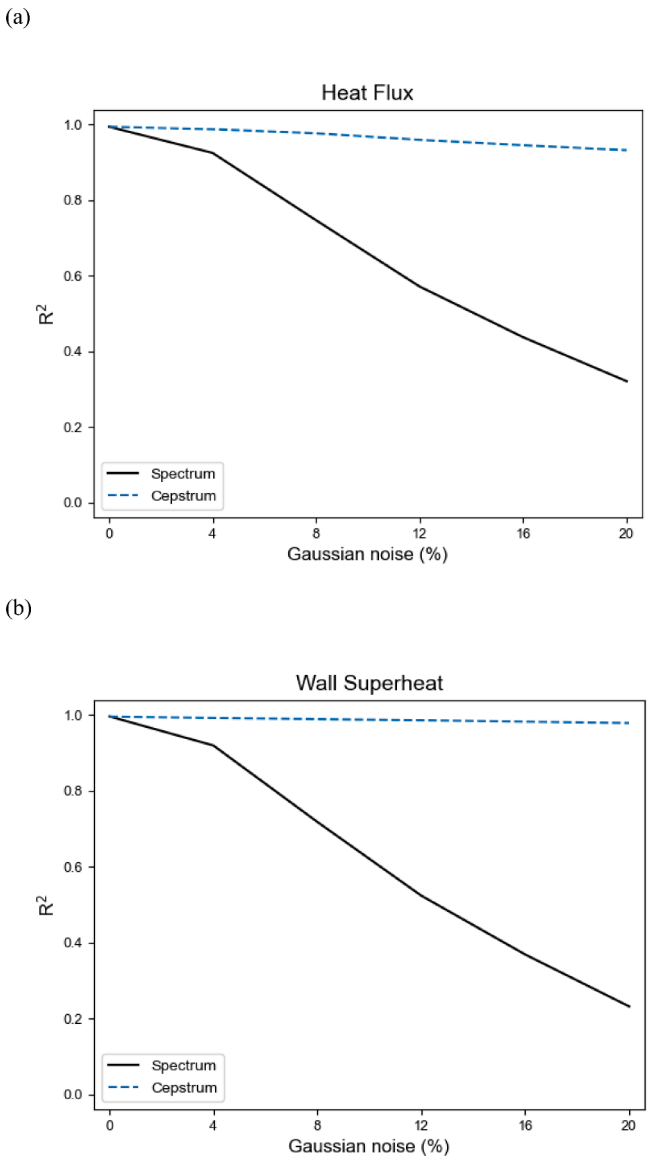


Fig. 13. Coefficient of determination (R^2) for the DNN regression model with Gaussian noise at $\Delta T_{sub} = 50$ K: (a) the case of heat-flux and (b) wall-superheat.

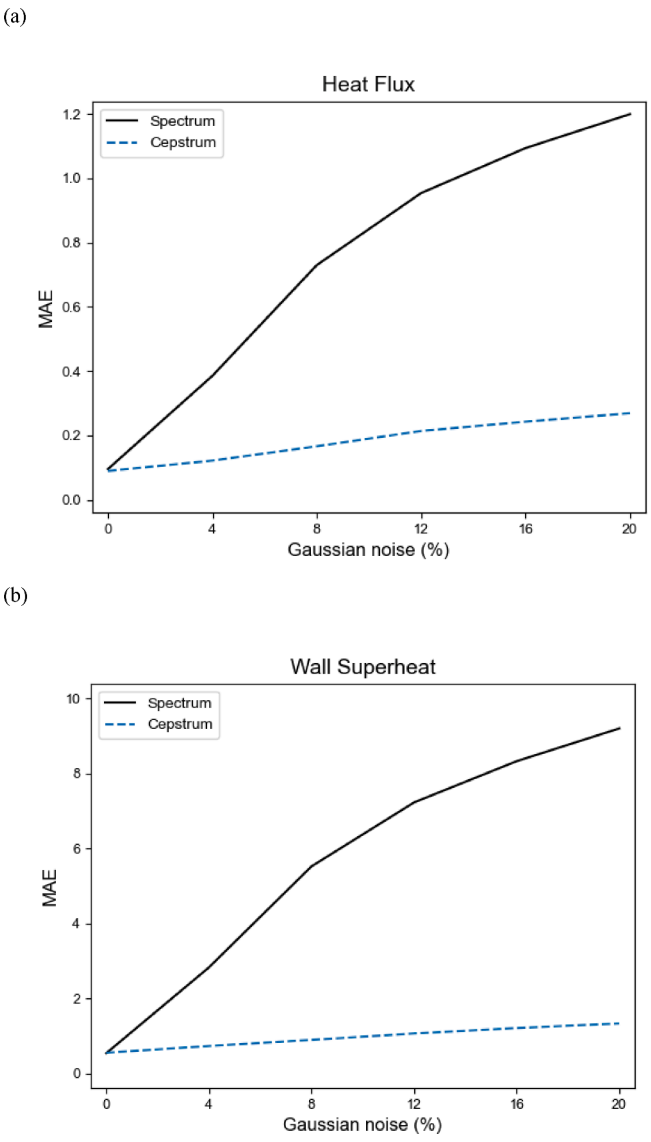


Fig. 14. Mean absolute error (MAE) for the DNN regression model with Gaussian noise at $\Delta T_{sub} = 50$ K: (a) the case of heat-flux and (b) wall-superheat.

changes in the features of the spectra. We speculate that the influence was reduced in the model based on cepstrum data because the Gaussian noise led to small changes in the spectra. The results showed that using cepstrum data for the preprocessing of the training data of an ML model is effective when predicting a boiling state using boiling sound.

CRediT authorship contribution statement

Junichiro Ono: Data curation, Visualization, Methodology. **Yuta Aoki:** Investigation, Visualization. **Noriyuki Unno:** Methodology, Validation. **Kazuhisa Yuki:** Methodology, Validation. **Koichi Suzuki:** Methodology. **Yoshitaka Ueki:** Methodology. **Shin-ichi Satake:** Conceptualization, Writing – original draft, Supervision, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Data availability

No data was used for the research described in the article.

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